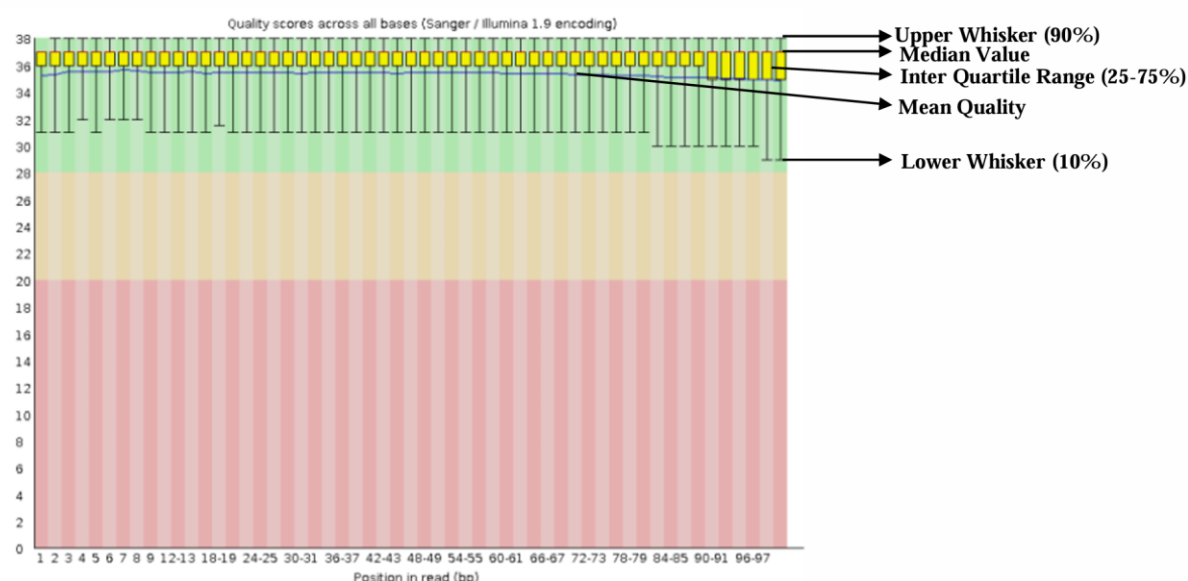


Results:

The controls and the three drugs (EPZ, UNC and GSK) were analysed. The initial analysis was to check the quality of the samples taken. The quality control analysis were performed using FASTQC.

Overall, both the controls and the three drugs had both forward and reverse strands with total bases of 2.5 Gbp. None of the sequences were flagged as poor quality strands. Each strand had a sequence length of 100 bp. The percentage of GC content the strands were found to 50 in the controls and the drugs EPZ and UNC, whereas GSK had percentage of 49. The quality of the strands were determined with the following graphs,

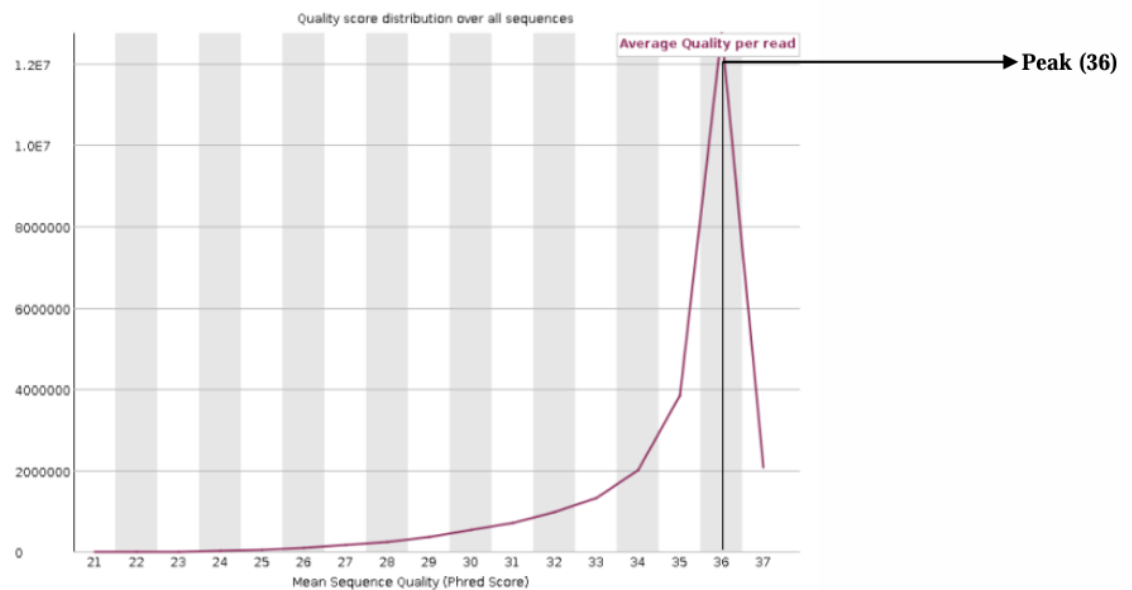
A) Per Base Sequence Quality Graph:



From the above graph, the median value was obtained to be 37 for strands of both control and the drugs. The inter-quartile range was found to be ranging between 35-37. The mean quality of the strands were found to be 35. The upper (90%) and lower (10%) whiskers were found to be 38 and in ranges between 28 to 32 for the controls and the drugs. The background indicates y-axis, the plots were found in the green region for all the sample data indicating very good quality score.

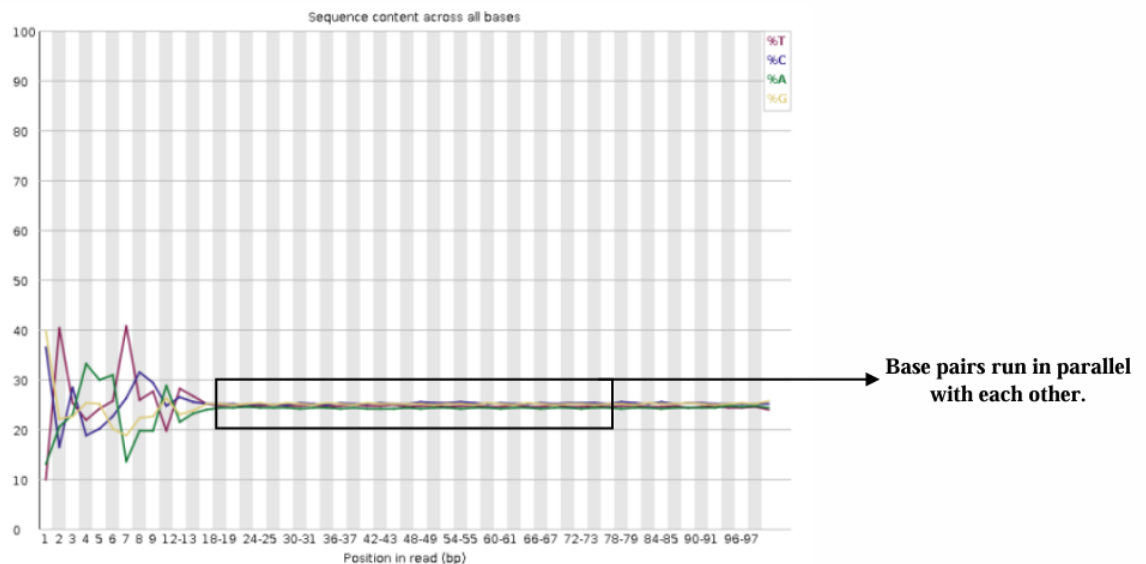
B) Per Sequence Quality Score:

The sequence quality score was obtained from the peak in the graph, indicating a phread score of 36 implying very good quality of the strands.



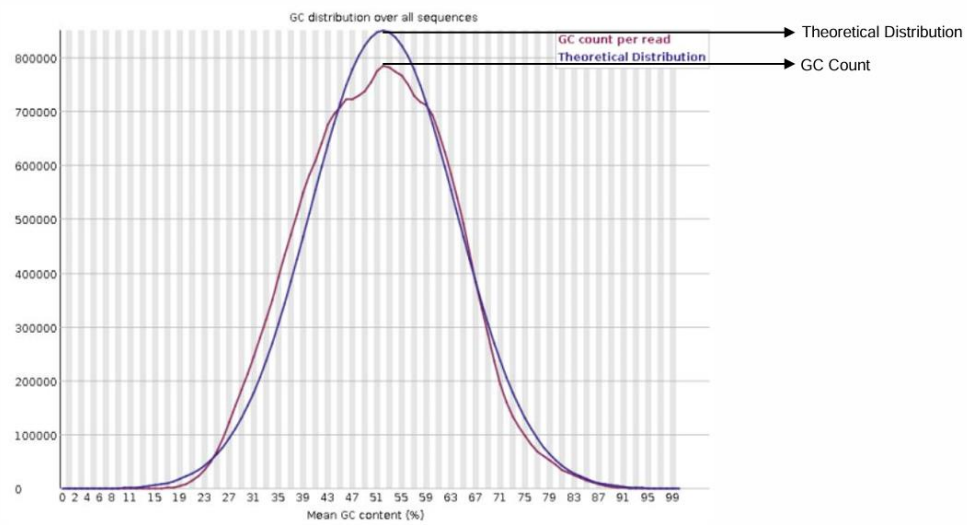
C) Per Base Sequence Content:

For the controls and the three drugs, the four bases (A,T,G and C) ranged between 10-40, distributed with no little difference as the samples were obtained from random library. The length of the reads were equal, %A = %T ; %G=%C and the plot line of the bases ran parallel to each other from the base position 18-19.



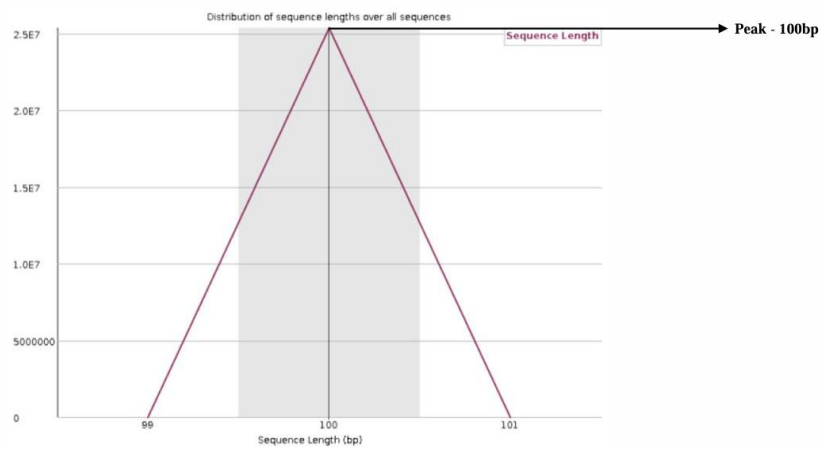
D) Per Sequence GC Content Graph:

For the controls and the three drugs, the theoretical distribution was 51 and the GC count per read peak was at 50 for the drugs EPZ and UNC and at 49 for GSK.



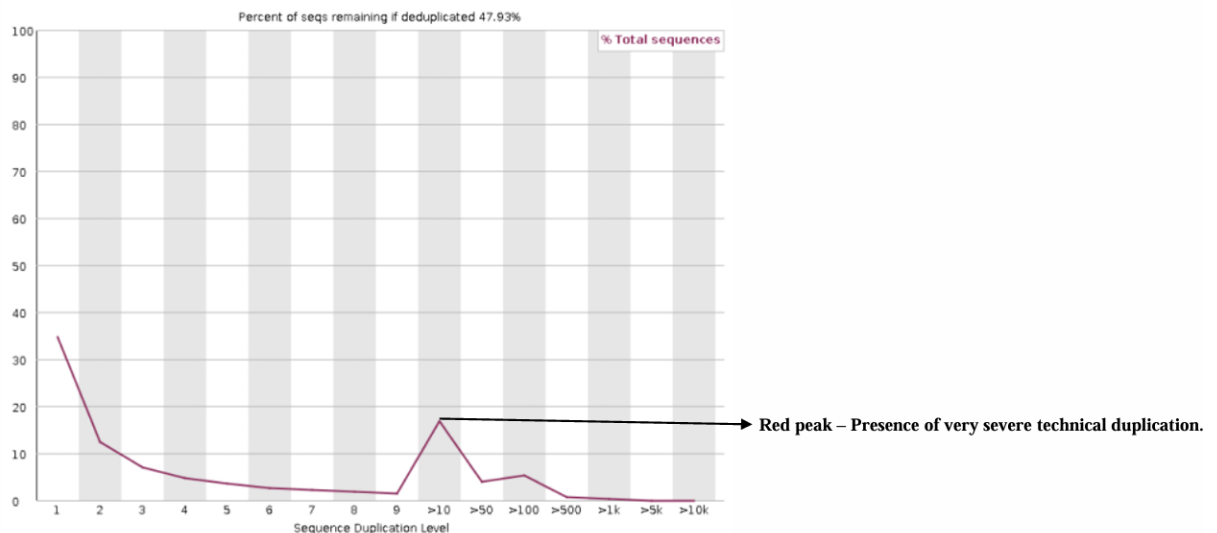
E) Sequence Length Distribution Graph:

For the controls and the three drugs, the peak was obtained at a sequence length of 100bp.



F) Sequence duplication Levels Graph:

For the controls and the three drugs, in the graph the blue line was absent indicating the absence of distribution of duplication levels for the full sequence set. The presence of the red line indicates the presence of the distribution of the de duplicated set derived from different duplication levels in the original data. The red peak was obtained at (>10) indicating presence of the large number of highly duplicated sequence. Thus, implying presence of very severe technical duplication.



HISAT2:

The alignment of the samples was obtained using Hisat2. The percentages of the paired and unpaired reads which were aligned concordantly and dis-concordantly were obtained. The overall alignment percentages were obtained for the controls ranging between 97.03 - 97.14, for the drugs, EPZ – 97.00 -97.12, UNC – 96.93- 97.00 and GSK – 96.29 - 96.42.

```
25686941 reads; of these:
25686941 (100.00%) were paired; of these:
1645664 (6.41%) aligned concordantly 0 times
22451232 (87.40%) aligned concordantly exactly 1 time
1590045 (6.19%) aligned concordantly >1 times
----
1645664 pairs aligned concordantly 0 times; of these:
251855 (15.30%) aligned discordantly 1 time
----
1393809 pairs aligned 0 times concordantly or discordantly; of these:
2787618 mates make up the pairs; of these:
1497905 (53.73%) aligned 0 times
1139371 (40.87%) aligned exactly 1 time
150342 (5.39%) aligned >1 times
97.08% overall alignment rate
```

The direction of the strands were obtained from the results of “Infer Experiment”

This is PairEnd Data

Fraction of reads failed to determine: 0.2616

Fraction of reads explained by "1++,1--,2+-,2-+": 0.3628

Fraction of reads explained by "1+-,1-+,2++,2--": 0.3756

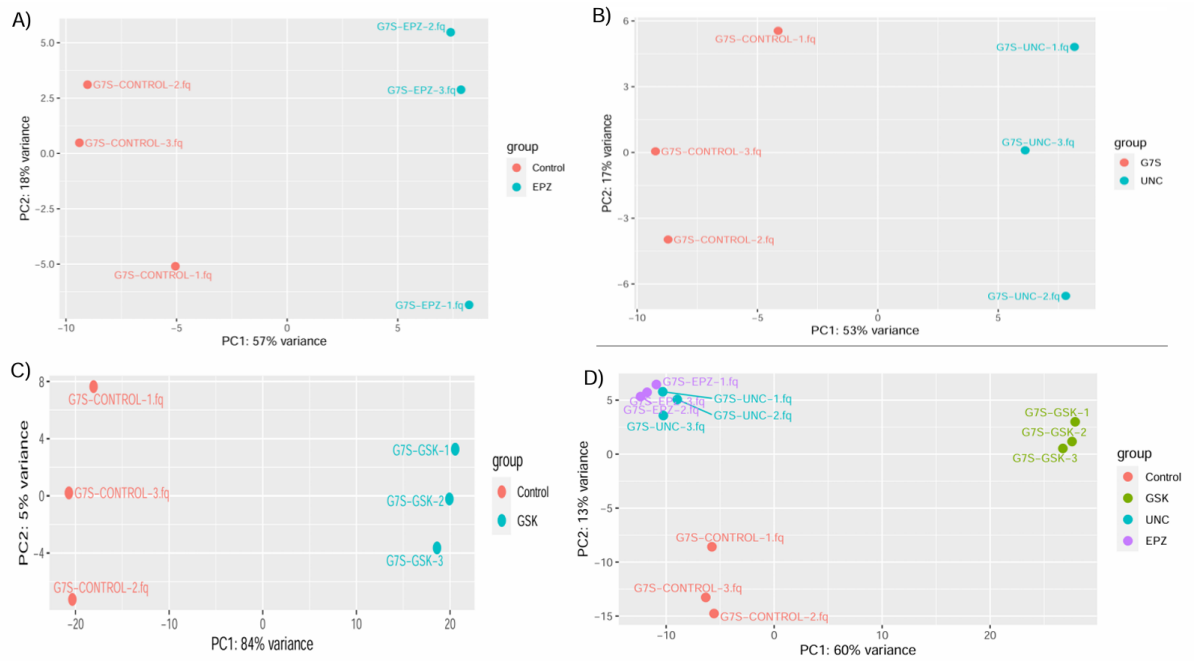
The fraction of reads failed to determine provides the strandness of the transcript that couldn't be determined. The fraction of reads explained by “ 1++, 1--,2+-, 2-+ “ when equally distributed for the forward and reverse strands, the strand is found to be un-stranded. The strands of controls and the three drugs EPZ, UNC and GSK were found to be un-stranded.

Deseq2

A) PCA :

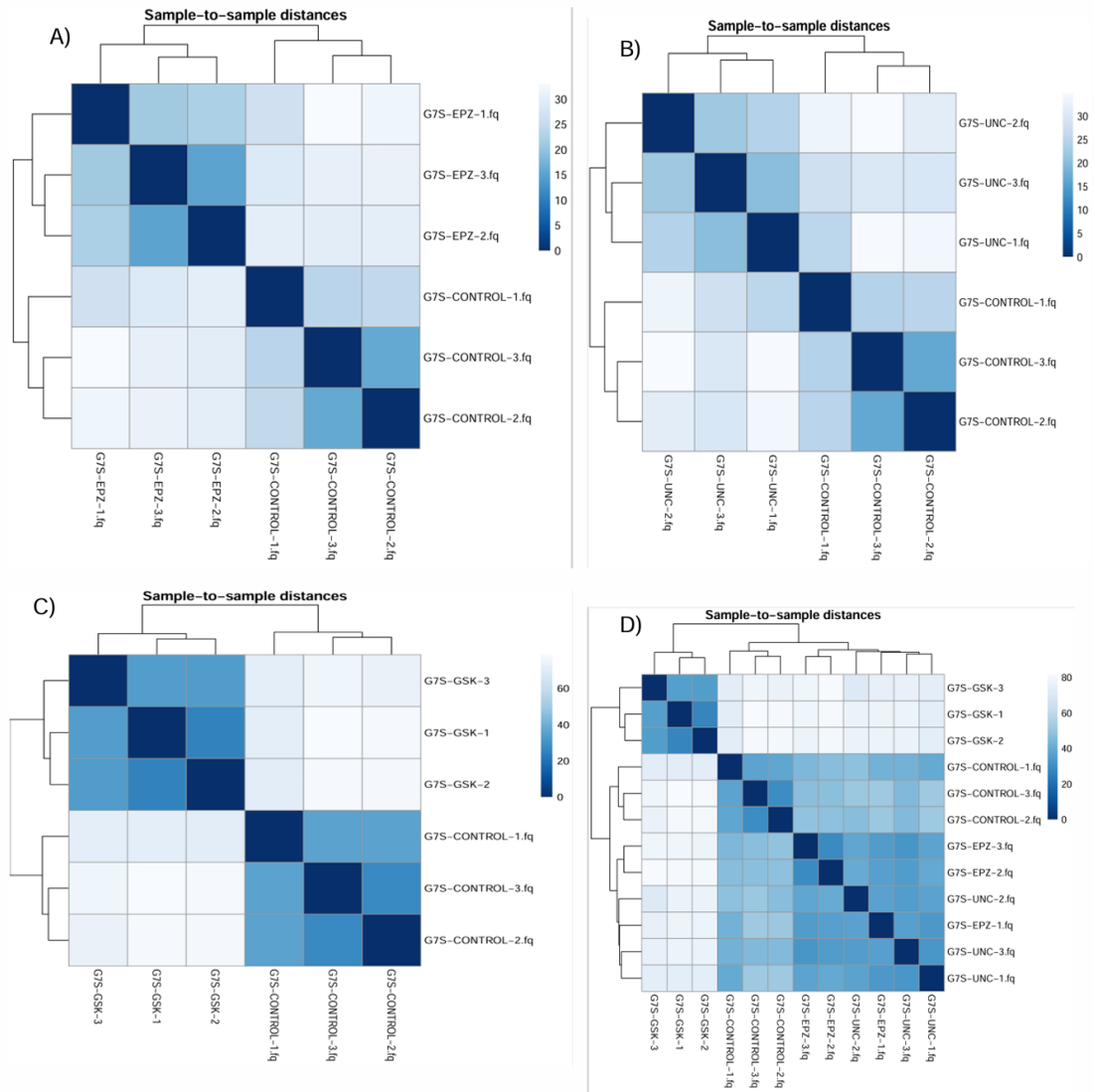
The PCA biplots were obtained with controls plotted against the three drugs.

(Fig A) Control vs EPZ: PC1 axis accounts for 57% variance in data, the EPZ triplicates were aligned in one direction whereas the control triplicates were aligned on the other. PC2 axis accounts 18%, EPZ 2 and 3 were aligned with Control 2 and 3 in the same direction whereas, EPZ 1 and Control 1 were aligned in the same direction. When components are grouped in circles they don't overlap each other. (Fig B) Control vs UNC: PC1 axis accounts for 53% variance in data, the UNC triplicates were aligned in one direction whereas the control triplicates were aligned on the other. PC2 axis accounts 17%, UNC 1 and 3 were aligned with Control 1 and 3 in the same direction whereas, UNC 2 and Control 2 were aligned in the same direction. When components are grouped in circles they don't overlap each other. (Fig C) Control Vs GSK: PC1 axis accounts for 84% variance in data, the GSK triplicates were aligned in one direction whereas the control triplicates were aligned on the other. PC2 axis accounts 5%, GSK 1 and 2 were aligned with Control 1 and 3 in the same direction whereas, GSK 3 and Control 2 were aligned in the same direction. When components are grouped in circles they don't overlap each other. (Fig D) Control Vs Drugs (EPZ vs UNC vs GSK) : PC1 axis accounts for 60% variance in data, the GSK triplicates were aligned in one direction whereas the control, EPZ and UNC triplicates were aligned on the other. PC2 axis accounts 13%, the triplicates of the drugs EPZ, UNC and GSK were aligned with in the same direction whereas, triplicates of the Control were aligned in the other direction. When components are grouped in circles control and GSK didn't overlap each other whereas EPZ and UNC were closely related.



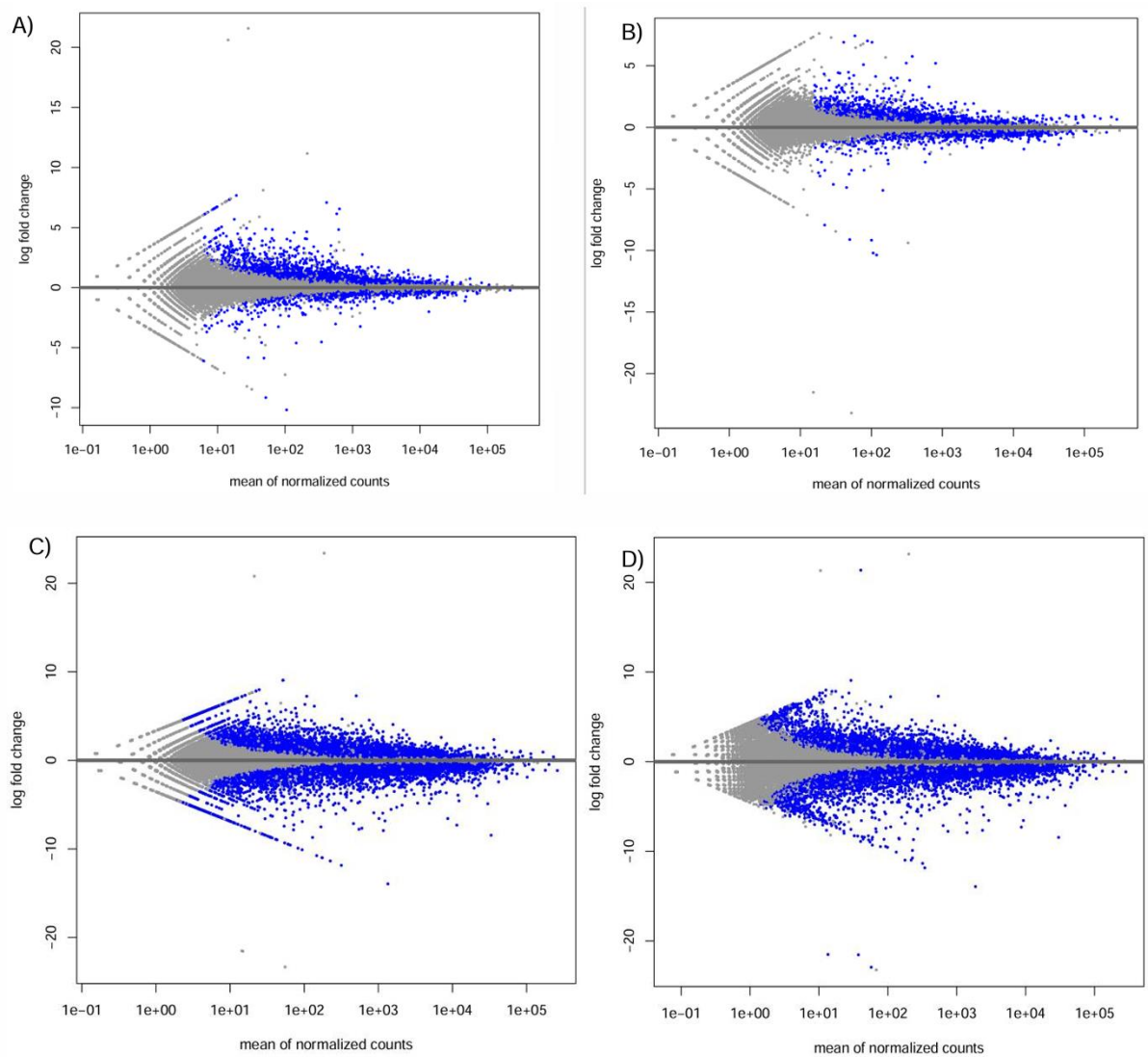
B) Heatmap: Hierarchical Clustering:

The similarities between the control and drugs were obtained from the heatmap, that sorts by similarities after hierarchical clustering and the sample distances are color coded. In all the graphs, the controls are clustered together and the drugs are clustered together. The diagonal line shaded in dark blue, indicates the sample distance as zero and self similar. The variance of shades from dark to light indicates the correlation. Darker shades are highly correlated whereas lighter shades are dissimilar. The samples within the same group occur together and form blue clusters. (Fig A) Control vs EPZ: The EPZ triplicates were correlated with itself and not highly with the triplicates of the control. (Fig B) Control vs UNC: The UNC triplicates were correlated with itself and not highly with the triplicates of the control. (Fig C) Control vs GSK : The GSK triplicates were correlated with itself and not highly with the triplicates of the control. (Fig D) Control Vs Drugs (EPZ vs UNC vs GSK) : The triplicates of the drugs EPZ and UNC were correlated, whereas GSK was found to be dissimilar.

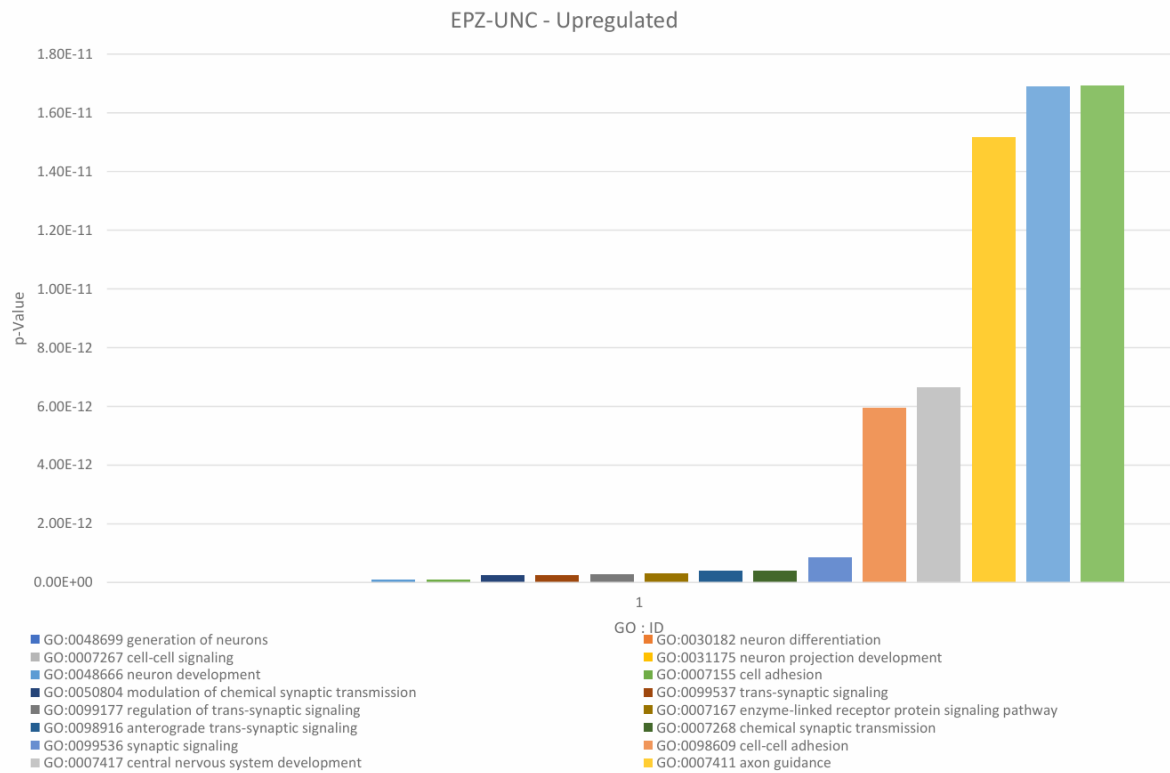
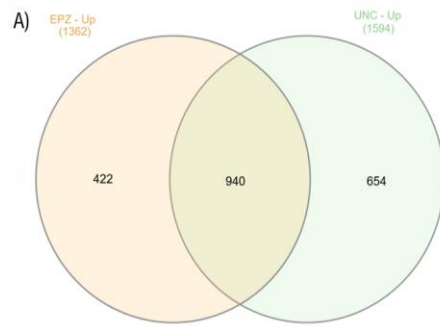


C) MA Plot:

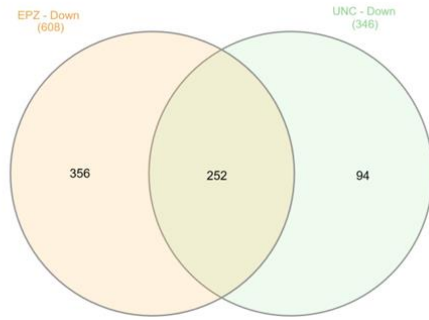
The Fold-change for each gene is plotted against mean of normalised count for the samples (control and three drugs). The scatter plot indicates differential expression of the genes in the sample. The blue dots which indicates presence of significant genes adjusted p-value <0.05. (Fig A) Control vs EPZ. (Fig B) Control vs UNC. (Fig C) Control vs GSK. (Fig D) Control vs Drugs (EPZ vs UNC vs GSK) In all graph, upper and lower blue dots on the left are further apart and closer towards right the sample is sequenced deeper or indicates presence of more replicates.



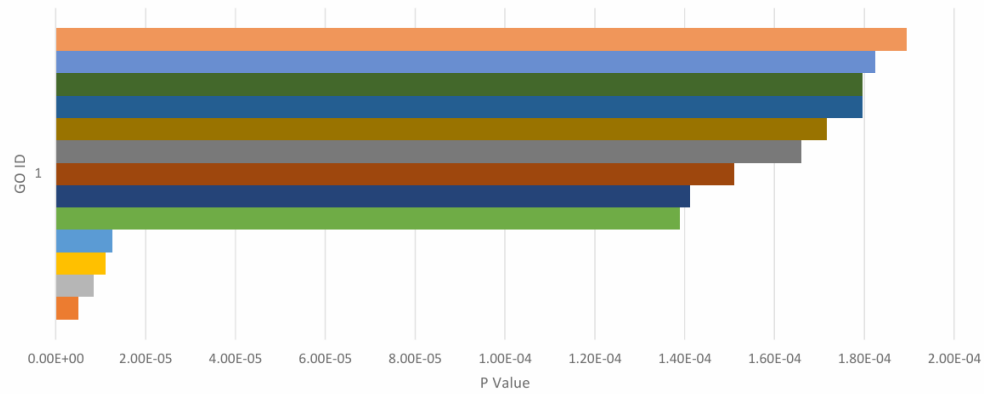
The differential expression of the genes were analysed using Topgene. The significant genes were found to be commonly present in both the drugs EPZ and UNC for both upregulated and downregulated genes. (Fig A) EPZ UNC Upregulated genes. (Fig B) EPZ UNC Downregulated genes. (Fig C) EPZ UNC GSK Upregulated genes. (Fig D) EPZ UNC GSK Downregulated genes.



B)

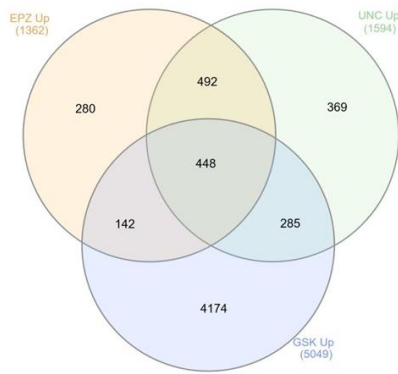


EPZ UNC Downregulated

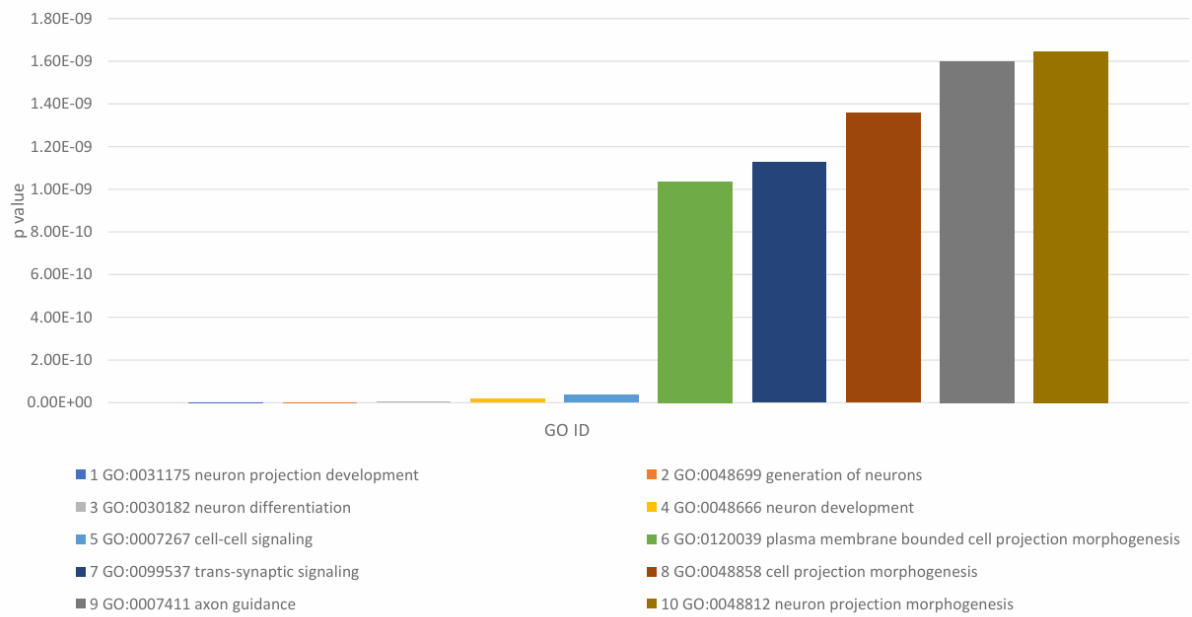


- GO:0071345 cellular response to cytokine stimulus
- GO:0010564 regulation of cell cycle process
- GO:0042790 nucleolar large rRNA transcription by RNA polymerase I
- GO:0030220 platelet formation
- GO:0034097 response to cytokine
- GO:0060218 hematopoietic stem cell differentiation
- GO:0035694 mitochondrial protein catabolic process
- GO:000902 cell morphogenesis
- GO:0051726 regulation of cell cycle
- GO:0016072 rRNA metabolic process
- GO:0009303 rRNA transcription
- GO:0048863 stem cell differentiation
- GO:0033043 regulation of organelle organization

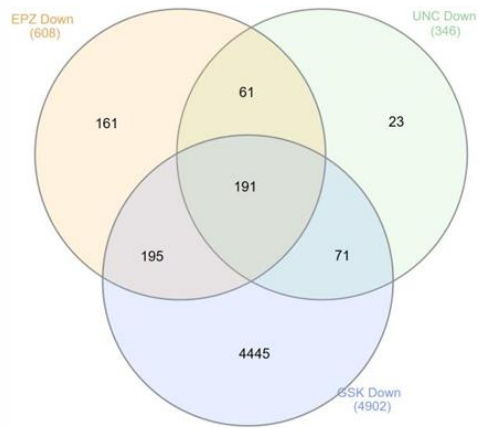
c)



EPZ - UNC - GSK - Upregulated



D)



EPZ UNC GSK Downregulated

